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10.5604/01.3001.0055.0638
Ximialis, P., Armatas, V., Scanlan, M., & Rachaniotis, N. P. (2025). The effects of situational variables on expected goals: A case study in international football. Journal of Kinesiology and Exercise Sciences, 35(109), 29-36. https://doi.org/10.5604/01.3001.0055.0638
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JKES
Journal of Kinesiology and Exercise Sciences

E-ISSN: 2956-4581
Journal of Kinesiology and Exercise Sciences

109 (35): 29-36, 2025 ORIGINAL RESEARCH

(1.3) DOI: 10.5604/01.3001.0055.0638

THE EFFECTS OF SITUATIONAL VARIABLES ON EXPECTED GOALS: A CASE STUDY IN INTERNATIONAL FOOTBALL

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Authors' contribution:

A. Study design/planning
B. Data collection/entry
C. Data analysis/statistics
D. Data interpretation
E. Preparation of manuscript
F. Literature analysis/search
G. Funds collection

Revised: 17.11.2024
Accepted: 17.02.2024
Ahead of Print: 27.03.2025
Published: 31.03.2025

Keywords: goal scoring, effectiveness, shots, match conditions, prediction

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Abstract:

Background: In the ever-evolving field of football analytics, understanding the factors influencing expected goals is crucial for gaining deeper insights into team performance. The aim of this study was to examine the effects of match conditions on the expected goals (xG) from defensive and offensive points of view in international elite football matches of the Greek national team.

Methods: Twenty-four (24) Greek national team matches during the period between 23 March 2019 and 14 November 2021 were analyzed for the study. In these selected matches the Greek national team took 369 shots and conceded 174 shots, out of which scored 26 goals, and conceded 23 goals to their opponents.

Results: Our findings suggest that situational variables such as time may influence expected goals (xG). More specifically, the Greek national team obtained higher expected goals for (xGf) during the second half of their matches compared to the first half. Zonal analysis of shots and goals revealed that central areas recorded more shots on goal in the analysis of both Greece and their opponents, but Greece favoured shooting from inside the penalty box more often than their opponents.

Conclusion: The findings emphasize the need for match analysts and coaches to consider the impact of situational variables during assessments of data analytics, especially in future xG research.

Introduction

Football is a relatively low-scoring sport and therefore unpredictable at match level, with singular and rare events such as a goal being scored having a much greater impact than in other team ball sports like basketball [1]. Therefore, it is common in football for the best-performing teams to sometimes lose a game, because although they may dominate in most measures of performance, their opponent may win the match by converting a single opportunity into a goal. This risk makes it difficult for coaches to make decisions during football matches, but in recent years, football analytics has attempted to improve decision-making and reduce the risk by providing data-driven analysis.

One of the most widespread and insightful metrics within football analytics is expected goals (xG), first introduced by Sam Green [2]. xG is a statistical model used to estimate the likelihood of a shot resulting in a goal. It is based on a range of contextual information at the time the shot was taken that can affect its outcome, such as the position of

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the shot, the angle of the shot, the type of pass that led to the shot, and the presence of defenders. This metric can be used as a measure of the offensive performance of a team or an individual player, and in its simplest form, this method involves calculating a team's chances of scoring and conceding goals.

As discussed above, the topic of xG is relatively new to the football community, but a vast number of studies have been undertaken to date. Lucey and colleagues [3] were the first to estimate goal probabilities using event and positional data in their model, using 10.000 shots of the English Premier League. Rathke [4] examined factors associated with predicting xG during English Premier League and Bundesliga matches, however, without taking the positions of defenders and goalkeepers into consideration. The results showed that the variables of distance and angle together significantly impacted xG and that xG can support forecasting future performance. Brechot and Flepp [5] proposed to use the xG models for performance evaluation instead of match outcomes, which may easily be influenced by randomness in short-term results. They introduced a chart built upon the concept of the xG by plotting the teams' ranking in the league table against their rankings based on xG. Bransen and Davis [6] investigated the differences between xG models trained on men and women's football event data. Results showed that although the xG models performed the same across the two environments, there were differences in feature importance between men's and women's games, as well as several in-game situations with xG differences, with women shooting from different positions than men and having a higher shot-to-goal conversion rate.

Despite the fact that the expected goals metric is a very active area of research and has been used in research and commercial settings, the model has come under fire [7-9]. This is due to (a) the number of factors that need to be included in calculating xG, (b) the lack of defensive positioning of players in relation to game-specific actions, and (c) not taking into account renowned clubs/players' performance during games. Although criticism is not without reason, the xG model is still seen as a valuable tool in predicting which players and teams will score and the likelihood of conceding goals [4].

Previous studies have identified that teams' performance can be influenced by situational variables such as match location, time, match outcome, team quality, and/or competition stage [10-12]. Yet, a vastly underexplored topic in the context of xG is how these models are affected by the situational variables during football matches. Thus, the aim of the present study was to examine the effects of match conditions on the expected goals (xG), from a defensive and offensive point of view, in international elite football matches.

Material and Methods

Sample

The sample consisted of n=24 Greek national team matches during the period between 23 March 2019 to 14 November 2021 in three different competitions, namely Euro 2020 Qualifiers (10 matches), Nations League 2021 (6 matches), and World Cup 2022 Qualifiers (8 matches). During these selected matches, the Greek national team took 369 and conceded 174 shots, out of which scored 26 and conceded 23 goals. Penalties were included in the study and assigned an xG value of 0.766 [13], which is the average conversion rate in the Bundesliga history. Free kicks and own goals were excluded.

The Regression Model

In order to calculate the expected goals for (xGf) and against (xGa) for the Greek national team, a logistic regression model was used that was first introduced by Shaw [14]:

$$xG = \beta 0 + \beta 1 \text{ x distance} + \beta 2 \text{ x angle} + \beta 3 \text{ x pressure} + \beta 4 \text{ x number of defenders.}$$

The regression model included four independent variables:

- **Distance:** the direct distance between the ball and the center of the goalpost measured automatically from Hudl Studio (Hudl, NE, USA), after calibrating the pitch size. The accuracy of this technique was determined by comparing the length between the penalty spot and the goal line to the actual dimension (11m), as proposed by Schulze and colleagues [15]. Accuracy was within previously estimated acceptable ranges for positional data [16], with an average absolute error of 0.8 ± 0.3%. The reliability of the above technique was tested by re-calculating 43 randomly selected shots (approximately 10% of the total) manually. By using the visible references of the goal, the penalty box, and the penalty spot, the perpendicular distance between the ball and the goal line (x) and the horizontal line between the ball and the center of the goal line (y) were calculated, as shown in Figure 1. Moreover, using the Pythagorean theorem, the direct distance between the ball and the center of the goal line was calculated:

$$z = \sqrt{x^2 + y^2}$$

The agreement between the automatic and manual measurements was tested with Cohen's kappa. The yielded kappa value of 0.85 reflects a strong agreement between the two measurements and thus acceptable reliability of the technique used in the present study.

- **Angle:** the angle formed by the two lines drawn from the two vertical posts was calculated (Figure 1), where 7.32m is the length of the goal. The result was converted into degrees as follows:

$$tan\Theta = \frac{7{,}32 * x}{x^2 + y^2 - \left(\frac{7{,}32}{2}\right)^2}$$

- **Pressure:** the defensive pressure exerted on the shot-taker, rated on a scale ranging from 0 to 5 [14].
- **Number of defenders:** the number of defensive players (including the goalkeeper) in a direct line from the shot position to the goal. In the example presented in Figure 1, the shot-taker confronted two defending players.

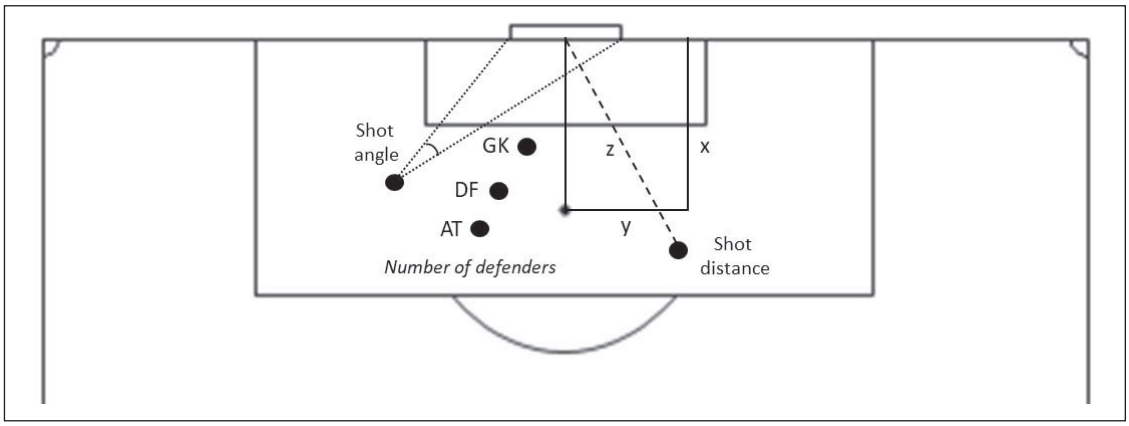

Figure 1. Definition and example of calculating shot distance, shot angle and number of defenders (AT: attacker, DF: defender, GK: goalkeeper)

Procedures

All shots for and against the Greek national team were observed in videos downloaded from Wyscout (Hudl, USA). An Excel (Microsoft, USA) spreadsheet was created to record the variables included in the model and the situational variables for each shot taken. The situational variables included:

1. **Match location**: the variable was recorded as "home" or "away", depending on whether the Greek national team was playing on its own ground or the opponent's [17].
2. **Time 45**: the variable was recorded as "1st half" or "2nd half", depending on the half that the sampled team's shot was taken [18].
3. **Time 15**: the playing time was divided into six periods of 15 minutes, i.e. "1-15", "16-30", "30-45+", "46-60", "61-75", "76-90+"; the variable recorded the period when the sampled team took the shot [18].
4. **Opponent quality**: the variable recorded the difference in the FIFA ranking between the competing teams [19]. Teams were separated into 2 groups based on the Greek national team's ranking (#45), published before the start of the Euro 2020 Qualifiers (released on 7 February 2019). Teams ranked higher than the Greek national team were categorized as "Group A"; teams ranked lower than the Greek national team were categorized as "Group B" [20].

Statistical Analysis

The statistical analysis was performed in SPSS Statistics for Windows, version 26.0 (SPSS Inc., IBM, USA), and the statistical significance was set at 5%. Prior to the analysis, the assumption of normality was verified using the Shapiro-Wilk test, while the assumption of homogeneity for xGf and xGa was verified using the Levene test. Paired t-test was used for the comparison of the groups referring to match location (Home, Away), opponent quality (Group A, Group B), and time 45 (1st Half, 2nd Half).

The mean number of expected goals for and against per match was compared among the six 15-minute periods using a series of non-parametric Kruskal-Wallis H tests, with p-values lower than 0.05 indicating significant time-

period effects. Whenever a significant time-period effect was found, individual pairs of variables (xGf and xGa) were compared between the two time-period groups using a series of Bonferroni-adjusted Mann-Whitney U tests, with p-values of under 0.004 (0.05/15) indicating significant differences. Effect sizes (Cohen's d) for between-period differences were determined, with values of 0.20, 0.50, and 0.80 representing small, moderate, and large differences, respectively [21].

Results

The results for xGf and xGa are presented in Table 1. A paired sample t-test was conducted to compare xGf between 1st Half and 2nd Half. There was a significant difference between the 2nd Half (mean=1.00, s.d.=0.82) and 1st Half (mean =0.69, s.d.=0.46). These results suggest that time does have an effect on xGf and specifically that the Greek national team obtained higher xGf during the second half of their matches compared to the first half. The remaining situational variables concerning xGf did not reveal any statistically significant differences in match location and opponent quality. However, it should be mentioned that there was a trend to obtain higher scores when playing at home and when playing against weaker teams from Group B.

The results for xGa revealed a significant difference in the scores when playing against teams from Group A (mean=1.60, sd=0.70) compared to Group B (mean=0.81, s.d.=0.59). Furthermore, there was a significant difference between the xGa in the 2nd Half (mean=0.64, s.d.=0.58) compared to the 1st Half (mean=0.39, s.d.=0.37). These results suggest that time and opponent quality do have an effect on xGa, i.e. the Greek national team suffered higher scores of xGa during the second half of their matches and when playing against a stronger opponent. No statistically significant differences were observed regarding match location.

Table 1. Statistics for expected goals for (xGf) and expected goals against (xGa)

	xGf (M±SD)	*t*-statistic	*p*-value	xGa (M±SD)	*t*-statistic	*p*-value
Home	1.97 ± 1.01	1.116	.288	1.12 ± 0.85	.557	.589
Away	1.43 ± 1.15			0.95 ± 0.59		
Group A	1.44 ± 0.95	-.290	.779	1.60 ± 0.70	3.358	.010*
Group B	1.56 ± 1.17			0.81 ± 0.59		
1st half	0.69 ± 0.46	-2.144	.041*	0.39 ± 0.37	-1.997	.047*
2nd half	1.00 ± 0.82			0.64 ± 0.58		

Note: *M* = Mean; *SD* = Standard deviation; * The mean difference is significant at the 0.05 level

The xGf and xGa for the 15-minute periods are presented in Figure 2. The Kruskal-Wallis H test showed a statistically significant difference in xGf between the six 15-minute periods (H(5) = 11.447, p = 0.043) but no significant differences were documented for xGa (H(5) = 6.317, p = 0.277). The post hoc Mann-Whitney U test revealed that xGf scores were significantly lower in the first 15-minute period (Md = 0.0650, n = 18) compared to the fourth 15-minute period (Md = 0.3050, n = 18), with U = 79.000, z = -2.647, p = 0.002, and a moderate effect size d = 0.62.

During the 24 matches studied, the Greek national team took 369 shots and scored 26 goals, while conceded 23

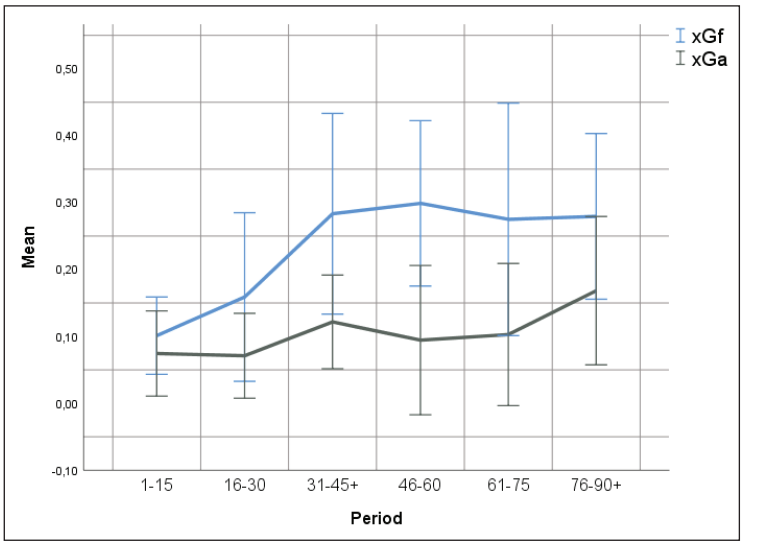

Figure 2. Expected goals for (xGf) and against (xGa) vs. 15-minute periods

goals from 174 shots. Concerning the taken shots as presented in Figure 3, the zone analysis revealed that the majority of shots were taken from the central zone located in front of the penalty box (33.3%). Moreover, the two zones inside the box, close to the penalty spot, were largely used. Overall, the central zones were used more frequently by the Greek national team compared to lateral zones, in terms of shooting and goal-scoring.

From a defensive point of view (Figure 4), the opponents used the two zones outside the penalty box more often and more specifically the central zone located just in front of the penalty box (42.5%). As mentioned above for the attacking shots, the central zones were used more frequently in relation to lateral zones from the opponent teams, both concerning shooting and goal-scoring.

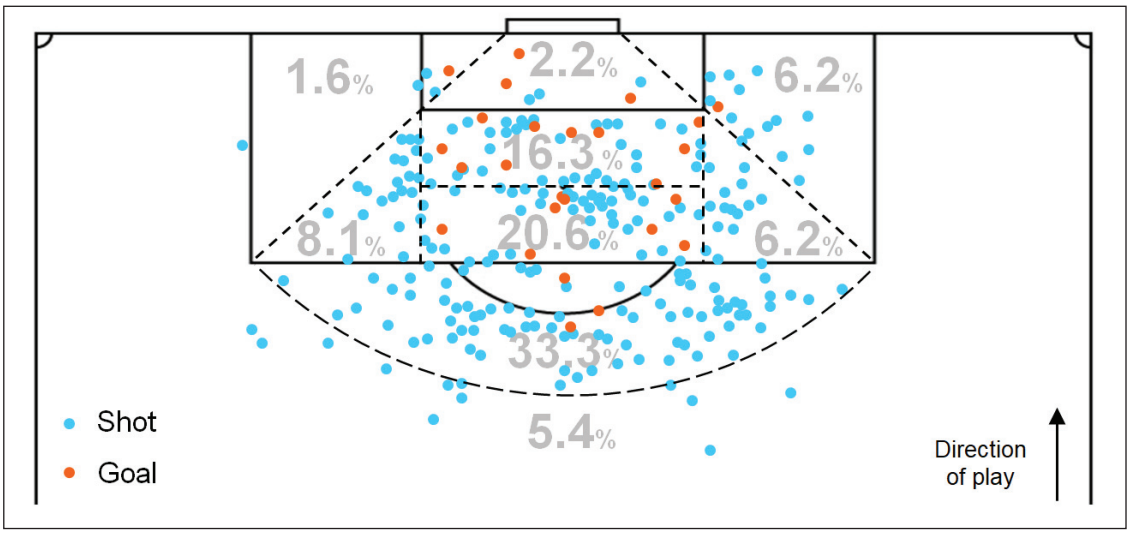

Figure 3. Scatterplot of shots and goal locations taken from the Greek national team

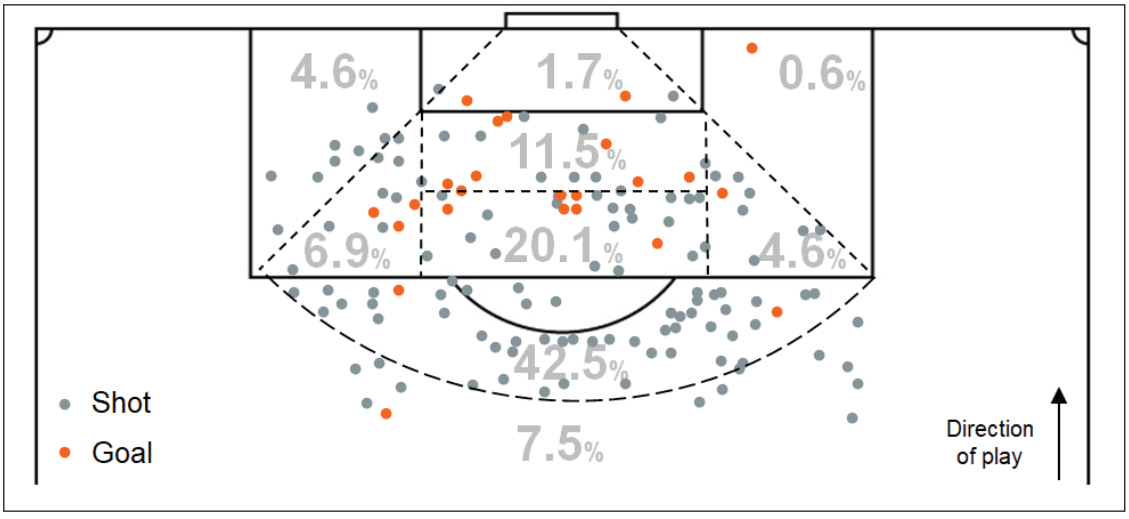

Figure 4. Scatterplot of shots and goal locations conceded by the Greek national team

Discussion

The aim of the present study was to examine the effects of match conditions on the expected goals from a defensive and offensive point of view in international elite football matches of the Greek national football team.

Results suggest that time does have an effect on xGf and more specifically that the Greek national team obtained higher xGf during the second half of their matches compared to the first half. As mentioned above, to the best of our knowledge, no studies have examined xG based on situational variables during football matches. However, a large number of studies attempted to identify goal-scoring patterns through the temporal analysis of scoring occurrences

and presented that a greater number of goals were scored in the second half of matches [18,22,23]. Moreover, Alberti and his colleagues [22] examined the English Premier League, the French Football League 1, the Italian Serie A, and the Spanish Football League and suggested that goal-scoring patterns are not related to particular seasons or country-related styles of play.

Moreover, the analysis revealed significant differences in xGa scores when playing against teams from Group A (mean=1.60, sd=0.70) compared to Group B (mean=0.81, s.d.=0.59). This indicates that the quality of the opponent had a notable impact on the defensive performance of the Greek national team. Furthermore, there was a statistically significant difference between xGa scores in the second half compared to the first half of matches. These results suggest that the Greek national team experienced higher scores of xGa during the second half of their matches. This finding suggests the possibility of fatigue, changes in defensive strategy, match context, or opponents' change in their attacking strategy as the match progresses. Importantly, the study found no statistically significant differences in xGa scores based on match location. This suggests that factors such as home advantage or environmental and atmospheric conditions did not have a significant impact on defensive performance in terms of xGa.

Concerning the analysis of time over a 15-minute period, a relationship between xGf and xGa and time of chance creation in the 24 matches analyzed for the study appeared, yet only xGf was found to be statistically significantly different when analyzed between the six 15-minute time periods. The findings of several studies support the contention that the majority of goals are scored late in each half, probably on account of fatigue, both physical [24,25] and mental [26], which accumulates as a match proceeds, leading to a greater number of technical failures [27]. Another reason could be the adoption of riskier attacking strategies in an attempt to change the current score [23,28]. It seems that the final fifteen minutes of the second half frequently represent a critical period in a match. Coaches should take this trend into account by reinforcing their team's defense, especially if the team does not enjoy a score advantage, or by making attacking substitutions or alterations to their teams' formation if they are looking to win games from a losing or drawing score position. The Kruskal-Wallis H test showed that there was a statistically significant difference in xGf for the Greek national team between the six 15-minute periods but no significant differences were found for xGa. The post hoc Mann-Whitney U test revealed that xGf scores were significantly lower in the first 15-minute period compared to the fourth 15-minute period, with a moderate effect size. Given the sample size for the study, it is advisable to continue this research across multiple teams for xGf and xGa, with a larger sample of matches in future study designs.

The remaining situational variables concerning xGf did not reveal any statistically significant differences in match location and opponent quality. However, it should be mentioned that there was a trend to obtain higher scores when playing at home and when playing against weaker teams from Group B.

With regards to the zone analysis for shots (for and against) and goals, the majority of shots on goal by Greece were taken from the central zone in front of the penalty box, accounting for 33.3% of the shots in total. The two zones inside the box, close to the penalty spot, between 6-18 yards away from the goal, were also heavily utilized. Overall, the Greek national team preferred the central zones for shooting and scoring goals, showing a higher frequency compared to the lateral zones. From a defensive standpoint (xGa), the opponents tended to use the two zones beyond the penalty spot more often (12-18 yards away), particularly the central zone just in front of the penalty box, which accounted for 42.5% of their shots. Similar to the attacking shots (xGf), the opponents also favored the central zones over the lateral zones for both shooting and scoring goals. Based on the number of shots taken (369) and conceded (174) across the 24 matches, it can be suggested that Greece adopted a proactive style of play that afforded them more than double the number of chances for (xGf) when compared to (xGa). However, we must observe that they only scored from ~7% (26/369) of their chances created, in comparison to ~13% (23/174) of shots from opponents resulting in goals. The discrepancy between the total number of chances for/against and goals scored could be because of the individual quality of the players taking the shot on goal, as well as the quality of goalkeepers involved in each match.

Limitations of study

There were many shortcomings in this study that should be noted. First, the analysis focused on 24 matches played by the Greek national team during a specific period. Expanding the dataset to include more matches across different competitions and involving other national teams as well as investigating xG patterns across various club teams with different playing styles and tactical approaches could strengthen the generalizability of the findings. Additionally, the study relied on publicly available data, which may have limitations in capturing the nuances of specific match events. Access to more comprehensive performance data could allow a more detailed analysis of the relationship between situational variables and xG. By acknowledging these limitations, we recognize the need for further research

to build upon this initial investigation. Future studies addressing these limitations can provide a more comprehensive understanding of how various match conditions interact with xG across different teams and playing styles in elite international football.

Conclusion

This study investigated the influence of match conditions on expected goals (xG) in international elite football, focusing on matches played by the Greek national team. The results revealed a clear trend: the Greek team generated a significantly higher xG value during the second half compared to the first, highlighting the importance of maintaining or improving attacking strategies later in the game. Additionally, the final 15 minutes of the second half emerged as a critical period for xG, suggesting potential fatigue or tactical adjustments impacting both teams. While the study did not find significant effects of match location on xG, there was a tendency for higher scores against weaker opponents, aligning with previous research. Both the Greek team and their opponents primarily targeted central areas for shots and goals, underlining the importance of controlling the central midfield and penalty area. The study emphasizes the need for future research to incorporate a wider range of situational variables and teams to offer even more comprehensive insights for coaches and analysts working in international elite football.

Practical implications

The above results could be practically incorporated into training to aid players' understanding and their needs, in particular how the offensive player takes a shot and how the defensive player should be positioned to defend this shot. The trend towards increased xGf in the final quarter of the second half suggests a potential window for tactical adjustments to maximize scoring opportunities. Coaches could consider introducing attacking substitutions or employing more aggressive tactics during this period to capitalize on this trend. The preference for central areas in both attacking and defending highlights the importance of controlling the central midfield and penalty area. Training exercises and tactical strategies can focus on maintaining possession and winning challenges in these crucial zones.

Conflict of interest: The authors certify that there is no conflict of interest with any financial organization regarding the material discussed in the manuscript.

Funding: This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Institutional Review Board Statement: All procedures were carried out in accordance with the 1964 Declaration of Helsinki and its subsequent amendments. Consent to perform testing was provided by the Bioethics Committee of the School of Physical Education and Sport Science, National and Kapodistrian University of Athens (1143/13-11-2019).

Data availability statement: The data presented in this study are available on request from the corresponding author.

Informed consent statement: The present study did not involve humans.

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Citation:
Ximialis P, Armatas V, Scanlan M, Rachaniotis NP: The effects of situational variables on expected goals: A case study in international football. Journal of Kinesiology and Exercise Sciences. 2025; 109 (35): 29-36.